AHM's

ANJUMAN INSTITUTE OF TECHNOLOGY AND MANAGEMENT

(Affiliated to Visvesvaraya Technological University, Belagavi and
Recognized by AICTE, New Delhi)
Bhatkal – 581320, Karnataka, India

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

WEB TECHNOLOGY LAB

BCSL504

SEMESTER: V

Academic Year: 2025-2026

SCHEME-2022

Name of the Student: …….

USN of the Student: ……..

Table of Contents

Sl. No	Title	Page. No
1.	Course Outcomes and Program Outcomes (POs)	**3-4**
2.	General Lab Guidelines	**5**
3.	Programming Exercises	**6-7**
4.	Assessment Details- CIE and SEE	**8-9**
5.	Maintaining Record and Execution Steps	**11**
6.	List of Experiments/Programs with Output	**12-53**

COURSE OUTCOMES

COURSE OUTCOMES	
At the end of the course the student will be able to:	
CO1	Design the experiment for the given problem using HTML, Javascript and CSS. .
CO2	Develop the solution for the given real-world problem using jQuery, Ajax and PHP.
CO3	Analyze the results and produce substantial written documentation.

COURSE LEARNING OBJECTIVES
• Learn HTML 5 elements and their use. • Use of CSS for enhanced user interface presentation • Gain knowledge of JavaScript, AJAX and jQuery for dynamic presentation. • Use of PHP to build Web applications. • Dessign and develop Websites and Web applications.

PROGRAM OUTCOMES (POs)

PROGRAM OUTCOMES/ GRADUAL ATTRIBUTES	
PO1	**Engineering Knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	**Problem Analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	**Design/Development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	**Conduct Investigations of Complex Problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	**Modern Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
PO6	**The Engineer and Society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	**Environment and Sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	**Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	**Individual and Team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	**Project Management and Finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	**Life-Long Learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

General Lab Guidelines

Do's

1. Maintain discipline in the laboratory.
2. Before entering the laboratory, keep your bags on the rack placed outside the lab.
3. Proper dress code with ID Cards is mandatory to get entry into the laboratory.
4. Students should carry a lab observation book, manual and record book must be up to date with the latest experiment written neatly.
5. Entry in the lab Utilization Register must be made before switching on the computer.
6. Maintain the same system throughout the course of the subject.
7. Program output with correct logic and program must be noted down in the lab observation book.
8. After completing the experiment, students must switch off the virtual machine first, followed by the main system, arrange the chair and leave the lab after getting the signature from the concerned faculty.

Don'ts

1. Do not come late to the laboratory.
2. Do not leave the laboratory without the permission of the faculty in-charge.
3. Never eat or drink while in the laboratory.
4. Do not alter computer/configuration settings.
5. Do not mishandle computers in the Laboratory.

Programming Exercises

Sl. No	Program
1	Develop the HTML page named as “Myfirstwebpage.html”. Add the following tags with relevant content. 1. Set the title of the page as “My First Web Page” 2. Within the body use the following tags: a) Moving text = “Basic HTML Tags” b) Different heading tags (h1 to h6) c) Paragraph d) Horizontal line e) Line Break f) Block Quote g) Pre tag h) Different Logical Style (<b>, <u>, <sub>, <sup> etc.)
2	Develop the HTML page named as “Table.html” to display your class time table. a) Provide the title as Time Table with table header and table footer, row-span and col-span etc. b) Provide various colour options to the cells (Highlight the lab hours and elective hours with different colours.) c) Provide colour options for rows.
3	Develop an external style sheet named as “style.css” and provide different styles for h2, h3, hr, p, div, span, time, img & a tags. Apply different CSS selectors for tags and demonstrate the significance of each.
4	Develop HTML page named as “registration.html” having variety of HTML input elements with background colors, table for alignment & provide font colors & size using CSS styles.
5	Develop HTML page named as “newpaper.html” having variety of HTML semantic elements with background colors, text-colors & size for figure, table, aside, section, article, header, footer… etc.
6	Apply HTML, CSS and JavaScript to design a simple calculator to perform the following operations: sum, product, difference, remainder, quotient, power, square-root and square
7	Develop JavaScript program (with HTML/CSS) for: a) Converting JSON text to JavaScript Object b) Convert JSON results into a date c) Converting From JSON To CSV and CSV to JSON d) Create hash from string using crypto.createHash() method
8	Develop a PHP program (with HTML/CSS) to keep track of the number of visitors visiting the web page and to display this count of visitors, with relevant headings. b. Develop a PHP program (with HTML/CSS) to sort the student records which are stored in the database using selection sort.

9	Develop jQuery script (with HTML/CSS) for: a. Appends the content at the end of the existing paragraph and list. b. Change the state of the element with CSS style using animate() method c. Change the color of any div that is animated.
10	Develop a JavaScript program with Ajax (with HTML/CSS) for: a. Use ajax() method (without Jquery) to add the text content from the text file by sending ajax request. b. Use ajax() method (with Jquery) to add the text content from the text file by sending ajax request. c. Illustrate the use of getJSON() method in jQuery d. Illustrate the use of parseJSON() method to display JSON values.

ASSESSMENT DETAILS (BOTH CIE AND SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

CONTINUOUS INTERNAL EVALUATION (CIE):

CIE marks for the practical course are 50 Marks.

The split-up of CIE marks for record/ journal and test are in the ratio 60:40.

● Each experiment is to be evaluated for conduction with an observation sheet and record write-up. Rubrics for the evaluation of the journal/write-up for hardware/software experiments are designed by the faculty who is handling the laboratory session and are made known to students at the beginning of the practical session.

● Record should contain all the specified experiments in the syllabus and each experiment write-up will be evaluated for 10 marks.

● Total marks scored by the students are scaled down to 30 marks (60% of maximum marks).

● Weightage to be given for neatness and submission of record/write-up on time.

● Department shall conduct a test of 100 marks after the completion of all the experiments listed in the syllabus.

● In a test, test write-up, conduction of experiment, acceptable result, and procedural knowledge will carry a weightage of 60% and the rest 40% for viva-voce.

● The suitable rubrics can be designed to evaluate each student's performance and learning ability.

● The marks scored shall be scaled down to 20 marks (40% of the maximum marks).

The Sum of scaled-down marks scored in the report write-up/journal and marks of a test is the total CIE marks scored by the student.

Semester End Evaluation (SEE):

● SEE marks for the practical course are 50 Marks.

● SEE shall be conducted jointly by the two examiners of the same institute, examiners are appointed by the Head of the Institute.

● The examination schedule and names of examiners are informed to the university before the conduction of the examination. These practical examinations are to be conducted between the schedule mentioned in the academic calendar of the University.

12082024Template for Practical Course and if AEC is a practical Course Annexure-V

● All laboratory experiments are to be included for practical examination.
● (Rubrics) Breakup of marks and the instructions printed on the cover page of the answer script to be strictly adhered to by the examiners. OR based on the course requirement evaluation rubrics shall be decided jointly by examiners.
● Students can pick one question (experiment) from the questions lot prepared by the examiners jointly.
● Evaluation of test write-up/ conduction procedure and result/viva will be conducted jointly by examiners.
● General rubrics suggested for SEE are mentioned here, writeup-20%, Conduction procedure and result in -60%, Viva-voce 20% of maximum marks. SEE for practical shall be evaluated for 100 marks and scored marks shall be scaled down to 50 marks (however, based on course type, rubrics shall be decided by the examiners)
Change of experiment is allowed only once and 15% of Marks allotted to the procedure part are to be made zero.
The minimum duration of SEE is 02 hours

RULES FOR MAINTAINING LABORATORY RECORD

1. Write your name, USN, Branch and Semester with the correct course name and course code on the outside front cover of the record. Write that same information on the first page inside.
2. Update Table of Contents every time you start each new experiment or topic
3. Always use black pen and write neatly and clearly.
4. Date to be written on the right-side page on the top right side corner.
5. On right-side page
   →Title of experiment
   →Aim/Objectives
   →Program
6. On left side page
   →Algorithm (if necessary)
   →Results

Sl. No	Name of the Experiment	Page No.
1.	Myfirstwebpage	**13-14**
2.	TimeTable	**16-18**
3.	Styling html using css	**19-21**
4.	Registration	**22-24**
5.	Newpaper	**25-31**
6.	Simple calculator	**32-35**
7.	a. Convert json text to javascript object b. Convert json result to date c. Convert json to csv & csv to json d. Create hash from string using crypto.createHash() method	**36-41**
8.	a. Track the number of visitor visiting webpage b. Sort the student record using selection sort	**42-46**
9.	a. Appends the content at the end of the existing paragraph and list. b. Change the state of the element with CSS style using animate() method c. Change the color of any div that is animated.	**47-49**
10.	a. Use ajax() method (without Jquery) to add the text content from the text file by sending ajax request. b. Use ajax() method (with Jquery) to add the text content from the text file by sending ajax request. c. Use of getJSON() method in jQuery d. Use of parseJSON() method to display JSON values.	**50-53**

Program 1

Develop the HTML page named as "Myfirstwebpage.html". Add the following tags with relevant content.

1. Set the title of the page as "My First Web Page"

2. Within the body use the following tags:

a) Moving text = "Basic HTML Tags"

b) Different heading tags (h1 to h6)

c) Paragraph

d) Horizontal line

e) Line Break

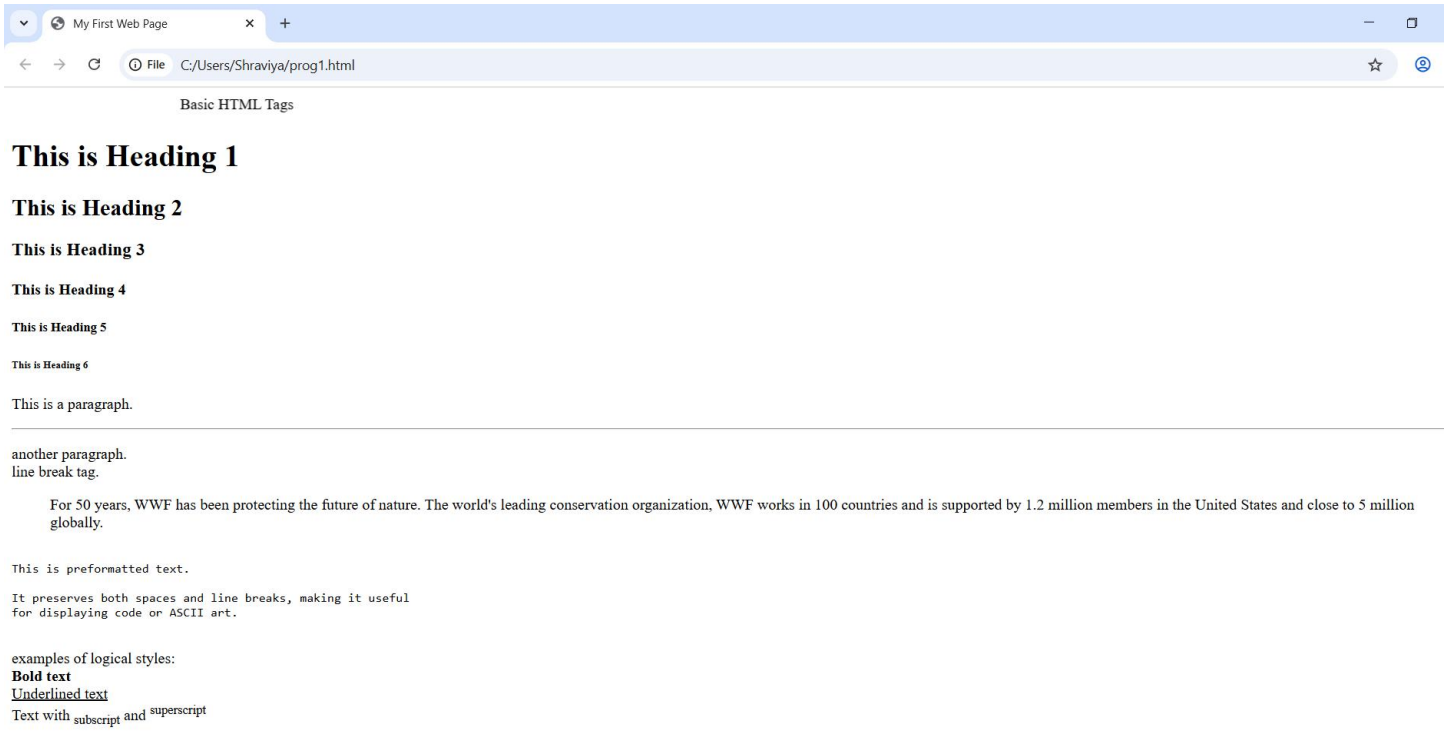
f) Block Quote

g) Pre tag

h) Different Logical Style (<b>, <u>, <sub>, <sup> etc.)

```
<!DOCTYPE html>
<head>
<title>My First Web Page</title>
</head>
<body>
<marquee>Basic HTML Tags</marquee>
<h1>This is Heading 1</h1>
<h2>This is Heading 2</h2>
<h3>This is Heading 3</h3>
<h4>This is Heading 4</h4>
<h5>This is Heading 5</h5>
<h6>This is Heading 6</h6>
<p>This is a paragraph.</p>
<hr>
<p>another paragraph.<br>line break tag.</p>
<blockquote cite="http://www.worldwildlife.org/who/index.html">
For 50 years, WWF has been protecting the future of nature. The world's leading conservation organization, WWF works in
100 countries and is supported by 1.2 million members in the United States and close to 5 million globally.
</blockquote>
<pre>
This is preformatted text.
It preserves both spaces and line breaks, making it useful
for displaying code
</pre>
<p>
examples of logical styles:<br>
<b>Bold text</b><br>
<u>Underlined text</u><br>
Text with <sub>subscript</sub> and <sup>superscript</sup>
</p>
</body>
</html>
```

Output:


My First Web Page
File
C:/Users/Shraviya/prog1.html
Basic HTML Tags
This is Heading 1
This is Heading 2
This is Heading 3
This is Heading 4
This is Heading 5
This is Heading 6
This is a paragraph.
another paragraph.
line break tag.
For 50 years, WWF has been protecting the future of nature. The world's leading conservation organization, WWF works in 100 countries and is supported by 1.2 million members in the United States and close to 5 million globally.
This is preformatted text.
It preserves both spaces and line breaks, making it useful
for displaying code or ASCII art.
examples of logical styles:
Bold text
Underlined text
Text with subscript and superscript

Program 2.

Develop the HTML page named as "Table.html" to display your class time table.
a) Provide the title as Time Table with table header and table footer, row-span and col-span etc.
b) Provide various colour options to the cells (Highlight the lab hours and elective hours with different colours.)
c) Provide colour options for rows.

```
<!DOCTYPE html>
<head>
    <title>Time Table</title>
    <style>
        body {
            font-family: Arial, sans-serif;
        }
        table {
            width: 80%;
            margin: 20px auto;
            border-collapse: collapse;
        }
        th,
        td {
            border: 1px solid #ddd;
            padding: 8px;
            text-align: center;
        }
        th {
            background-color: #f4f4f4;
            color: #333;
        }
        tr:nth-child(even) {
            background-color: #f9f9f9;
        }
        tr:nth-child(odd) {
            background-color: #e6f7ff;
        }
        .lab-hour {
            background-color: #ffcccc;
        }
        .elective-hour {
            background-color: #ccffcc;
        }
        .highlight {
            font-weight: bold;
            color: #d63384;
        }
        tfoot {
            background-color: #e0e0e0;
            font-weight: bold;
        }
    </style>
</head>
<body>
    <h1 style="text-align: center;">Time Table</h1>
    <table>
        <thead>
            <tr>
                <th>Day/Time</th>
                <th>9:00 - 10:00</th>
                <th>10:00 - 11:00</th>
                <th>11:00 - 12:00</th>
```

```
                <th>12:00 - 1:00</th>
                <th>Lunch Break</th>
                <th>2:00 - 3:00</th>
                <th>3:00 - 4:00</th>
            </tr>
        </thead>
        <tbody>
            <tr>
                <td>Monday</td>
                <td>Math</td>
                <td>English</td>
                <td class="lab-hour">Physics Lab</td>
                <td>Elective</td>
                <td rowspan="5" class="highlight">Break</td>
                <td class="elective-hour">Elective</td>
                <td>History</td>
            </tr>
            <tr>
                <td>Tuesday</td>
                <td class="elective-hour">Elective</td>
                <td>Biology</td>
                <td>Math</td>
                <td class="lab-hour">Chemistry Lab</td>
                <td>Geography</td>
                <td>PE</td>
            </tr>
            <tr>
                <td>Wednesday</td>
                <td>History</td>
                <td class="lab-hour">Computer Lab</td>
                <td>English</td>
                <td>Math</td>
                <td>Physics</td>
                <td class="elective-hour">Elective</td>
            </tr>
            <tr>
                <td>Thursday</td>
                <td>PE</td>
                <td>History</td>
                <td>Geography</td>
                <td class="elective-hour">Elective</td>
                <td>Biology</td>
                <td>Math</td>
            </tr>
            <tr>
                <td>Friday</td>
                <td class="lab-hour">Biology Lab</td>
                <td>Math</td>
                <td>English</td>
                <td>Physics</td>
                <td class="elective-hour">Elective</td>
                <td>Chemistry</td>
            </tr>
        </tbody>
        <tfoot>
            <tr>
                <td colspan="8">End of Timetable</td>
            </tr>
        </tfoot>
    </table>
</body>
</html>
```

Output:

Time Table

Day/Time	9:00 - 10:00	10:00 - 11:00	11:00 - 12:00	12:00 - 1:00	Lunch Break	2:00 - 3:00	3:00 - 4:00
Monday	Math	English	Physics Lab	Elective		Elective	History
Tuesday	Elective	Biology	Math	Chemistry Lab		Geography	PE
Wednesday	History	Computer Lab	English	Math	Break	Physics	Elective
Thursday	PE	History	Geography	Elective		Biology	Math
Friday	Biology Lab	Math	English	Physics		Elective	Chemistry
End of Timetable							

Program 3:

Develop an external style sheet named as "style.css" and provide different styles for h2, h3, hr, p, div, span, time, img & a tags. Apply different CSS selectors for tags and demonstrate the significance of each.

HTML

```html
<!DOCTYPE html>
<head>
<title>Sample Styled Page (No Div)</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<main id="main-content">
<h2>Welcome to Our Styled Page</h2>
<p>This is a paragraph right after an h2. It demonstrates the adjacent sibling selector.</p>
<h3>Hover over me!</h3>
<hr>
<p>Here's a <span class="highlight">highlighted</span> word using the class selector.</p>
<section>
<p>This paragraph is inside a section, showing the descendant selector.</p>
<span>This span is a direct child of the section.</span>
</section>
<p>The current date and time: <time datetime="2023-08-31">August 31, 2023</time></p>
<p>Notice how the first letter of each paragraph is styled differently.</p>
<article class="special">
<p>This paragraph is inside an article with class="special".</p>
</article>
<img src="https://via.placeholder.com/150" alt="A placeholder image">
<p>Check out this <a href="https://www.aitm.edu.in">link</a> to see different link states.</p>
</main>
</body>
</html>
```

CSS

```css
h2 {
color: #2c3e50;
font-family: 'Arial', sans-serif; border-bottom: 2px solid #3498db; padding-bottom: 10px;
}
h3:hover {
color: #e74c3c; cursor: pointer;
transition: color 0.3s ease;
}
hr {
border: 0; height: 1px;
background-image: linear-gradient(to right, rgba(0, 0, 0, 0), rgba(0, 0, 0,
0.75), rgba(0, 0, 0, 0));
}
p[lang] {
font-style: italic;
}
.highlight {
background-color: #f1c40f; padding: 5px;
}
#main-content {
max-width: 800px; margin: 0 auto; padding: 20px;
background-color: #ecf0f1;
}
div p {
line-height: 1.6; margin-bottom: 15px;
}
```

```
div > span {
font-weight: bold; color: #16a085;
}
h2 + p {
font-size: 1.1em; color: #7f8c8d;
}
time[datetime] {
color: #8e44ad; font-weight: bold;
}
p::first-letter {
font-size: 1.5em; font-weight: bold; color: #c0392b;
}
img, a {
border: 1px solid #bdc3c7; padding: 5px;
}
a:link, a:visited {
color: #3498db;
text-decoration: none;
}
a:hover, a:active { color: #e74c3c;
text-decoration: underline;
}
img[alt] {
max-width: 100%; height: auto;
}
div.special p {
text-indent: 20px; color: #27ae60;
}
```

Output:

Main Heading

Subheading

This is a paragraph demonstrating the basic text styling applied by CSS.

This is a styled **div** element with padding and a light border. Inside the div, we can also use **span elements** that have their own styles, like this bold and orange text.

Current time: 10:30 AM



Visit vtucode.in to learn more about our services.

This paragraph is highlighted with a yellow background.

This text is centered using a class selector.

This is a special paragraph with unique styles applied through an ID selector.

Program 4:

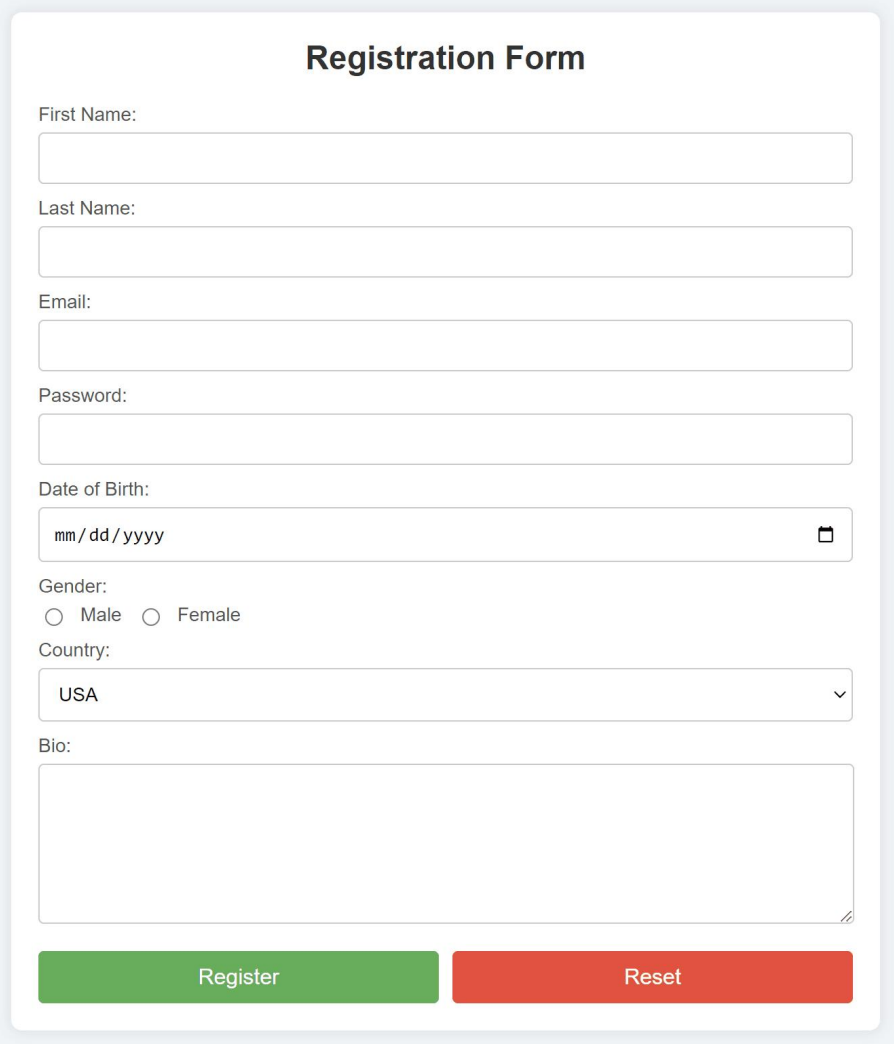
Develop HTML page named as "registration.html" having variety of HTML input elements with background colors, table for alignment & provide font colors & size using CSS styles.

```
><!DOCTYPE html>
<head>
    <title>Registration Form | vtucode</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f0f4f8;
            margin: 0;
            padding: 20px;
            display: flex;
            justify-content: center;
            align-items: center;
            min-height: 100vh;
        }
        .container {
            width: 100%;
            max-width: 600px;
            background-color: #fff;
            padding: 20px;
            border-radius: 8px;
            box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
            display: flex;
            flex-direction: column;
            gap: 20px;
        }
        h2 {
            text-align: center;
            color: #333;
            margin: 0;
        }
        .form-group {
            display: flex;
            flex-direction: column;
            gap: 5px;
            margin-bottom: 10px;
        }
        label {
            font-size: 14px;
            color: #555;
        }
        input[type="text"],
        input[type="email"],
        input[type="password"],
        input[type="date"],
        select,
        textarea {
            padding: 10px;
            border: 1px solid #ccc;
            border-radius: 4px;
            font-size: 14px;
        }
        .gender-options {
            display: flex;
            gap: 10px;
            align-items: center;
        }
        input[type="submit"],
```

```
        input[type="reset"] {
            padding: 10px 20px;
            border: none;
            border-radius: 4px;
            cursor: pointer;
            font-size: 16px;
            flex: 1;
        }
        .button-group {
            display: flex;
            gap: 10px;
            justify-content: center;
        }
        input[type="submit"] {
            background-color: #4CAF50;
            color: white;
        }
        input[type="reset"] {
            background-color: #f44336;
            color: white;
        }
        .form-group textarea {
            margin-bottom: 10px;
        }
    </style>
</head>
<body>
    <div class="container">
        <h2>Registration Form</h2>
        <form action="#" method="post">
            <div class="form-group">
                <label for="firstName">First Name:</label>
                <input type="text" id="firstName" name="firstName" required>
            </div>
            <div class="form-group">
                <label for="lastName">Last Name:</label>
                <input type="text" id="lastName" name="lastName" required>
            </div>
            <div class="form-group">
                <label for="email">Email:</label>
                <input type="email" id="email" name="email" required>
            </div>
            <div class="form-group">
                <label for="password">Password:</label>
                <input type="password" id="password" name="password" required>
            </div>
            <div class="form-group">
                <label for="dob">Date of Birth:</label>
                <input type="date" id="dob" name="dob">
            </div>
            <div class="form-group">
                <label>Gender:</label>
                <div class="gender-options">
                    <input type="radio" id="male" name="gender" value="male">
                    <label for="male">Male</label>
                    <input type="radio" id="female" name="gender" value="female">
                    <label for="female">Female</label>
                </div>
            </div>
            <div class="form-group">
                <label for="country">Country:</label>
                <select id="country" name="country">
```

```
                <option value="usa">USA</option>
                <option value="canada">Canada</option>
                <option value="uk">UK</option>
                <option value="india">India</option>
            </select>
        </div>
        <div class="form-group">
            <label for="bio">Bio:</label>
            <textarea id="bio" name="bio" rows="4"></textarea>
        </div>
        <div class="button-group">
            <input type="submit" value="Register">
            <input type="reset" value="Reset">
        </div>
    </form>
  </div>
</body>
</html>
```

Output:



Program 5:

Develop HTML page named as "newpaper.html" having variety of HTML semantic elements with background colors, text-colors & size for figure, table, aside, section, article, header, footer… etc.

```
<!DOCTYPE html>
<head>
    <title>Newspaper Page | vtucode</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
        body {
            padding: 20px;
            font-family: 'Arial', sans-serif;
            color: #000000;
            display: flex;
            flex-direction: column;
            min-height: 100vh;
        }
        header {
            margin-bottom: 30px;
            border-radius: 10px;
            align-items: center;
            background-color: #7b38f7;
            color: #fff;
            padding: 20px;
            display: flex;
            justify-content: space-between;
            text-align: center;
            box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
        }
        header a {
            font-size: 25px;
            font-weight: 600;
            color: #fff;
            text-decoration: none;
        }
        nav {
            display: flex;
            gap: 20px;
            color: #fff;
            text-align: center;
        }
        nav a {
            font-size: 18px;
            color: #fff;
            text-decoration: none;
            font-weight: bold;
        }
        nav a:hover {
            text-decoration: underline;
        }
        .content {
            display: flex;
            justify-content: space-between;
            flex: 1;
            margin: auto;
            padding: 20px 0;
```

```
        gap: 20px;
        position: relative;
    }
    .main-content {
        cursor: pointer;
        flex: 1;
        display: grid;
        grid-template-columns: repeat(auto-fill, minmax(300px, 1fr));
        gap: 20px;
        background-color: #fff;
        border-radius: 12px;
        padding: 25px;
        box-shadow: rgba(9, 30, 66, 0.25) 0px 1px 1px, rgba(9, 30, 66, 0.13) 0px 0px 1px 1px;
    }
    aside {
        border: 1px solid #ccc;
        padding: 20px;
        width: 350px;
        border-radius: 8px;
        box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
        position: -webkit-sticky;
        position: sticky;
        top: 20px;
        color: #333;
        right: 0;
        margin-left: 20px;
    }
    .related-news h3 {
        text-align: center;
        border-radius: 7px;
        padding: 8px;
        background: #000;
        color: #ffffff;
        font-size: 1.4em;
        margin-bottom: 15px;
    }
    .related-news ul {
        list-style: outside;
        padding: 7px;
        margin: 0;
    }
    .related-news li {
        margin-bottom: 12px;
    }
    .related-news a {
        text-decoration: none;
        color: #7b38f7;
        font-weight: bold;
        transition: color 0.3s ease;
    }
    .related-news a:hover {
        text-decoration: underline;
    }
    footer {
        border-radius: 10px;
        background-color: #7b38f7;
        color: #fff;
        padding: 20px;
        font-weight: 500;
        text-align: center;
        margin-top: auto;
        font-size: 18px;
```

```
        }
        article {
            transition: all 0.3s ease;
            background-color: #fff;
            padding: 15px;
            box-shadow: rgba(9, 30, 66, 0.25) 0px 1px 1px, rgba(9, 30, 66, 0.13) 0px 0px 1px 1px;
            border-radius: 7px;
            color: #000000;
        }
        figure {
            background-color: #fafafa;
            padding: 10px;
            border: 1px solid #ddd;
            border-radius: 8px;
            text-align: center;
            margin: 0;
        }
        figcaption {
            font-size: 0.9em;
            color: #666;
        }
        img {
            max-width: 100%;
            height: auto;
            border-radius: 8px;
        }
        section {
            padding: 20px;
            width: 100%;
            background-color: #fff;
            border-radius: 8px;
            box-shadow: rgba(9, 30, 66, 0.25) 0px 1px 1px, rgba(9, 30, 66, 0.13) 0px 0px 1px 1px;
        }
        section h2 {
            color: #fff;
            background: #000;
            font-size: 24px;
            border-radius: 10px;
            text-align: center;
            padding: 10px;
            margin-bottom: 30px;
        }
        table {
            width: 100%;
            border-collapse: collapse;
        }
        caption {
            font-size: 18px;
            margin-bottom: 10px;
            color: #555;
        }
        thead {
            background-color: #007BFF;
            color: #fff;
        }
        th,
        td {
            padding: 12px;
            text-align: left;
        }
        th {
            font-weight: bold;
```

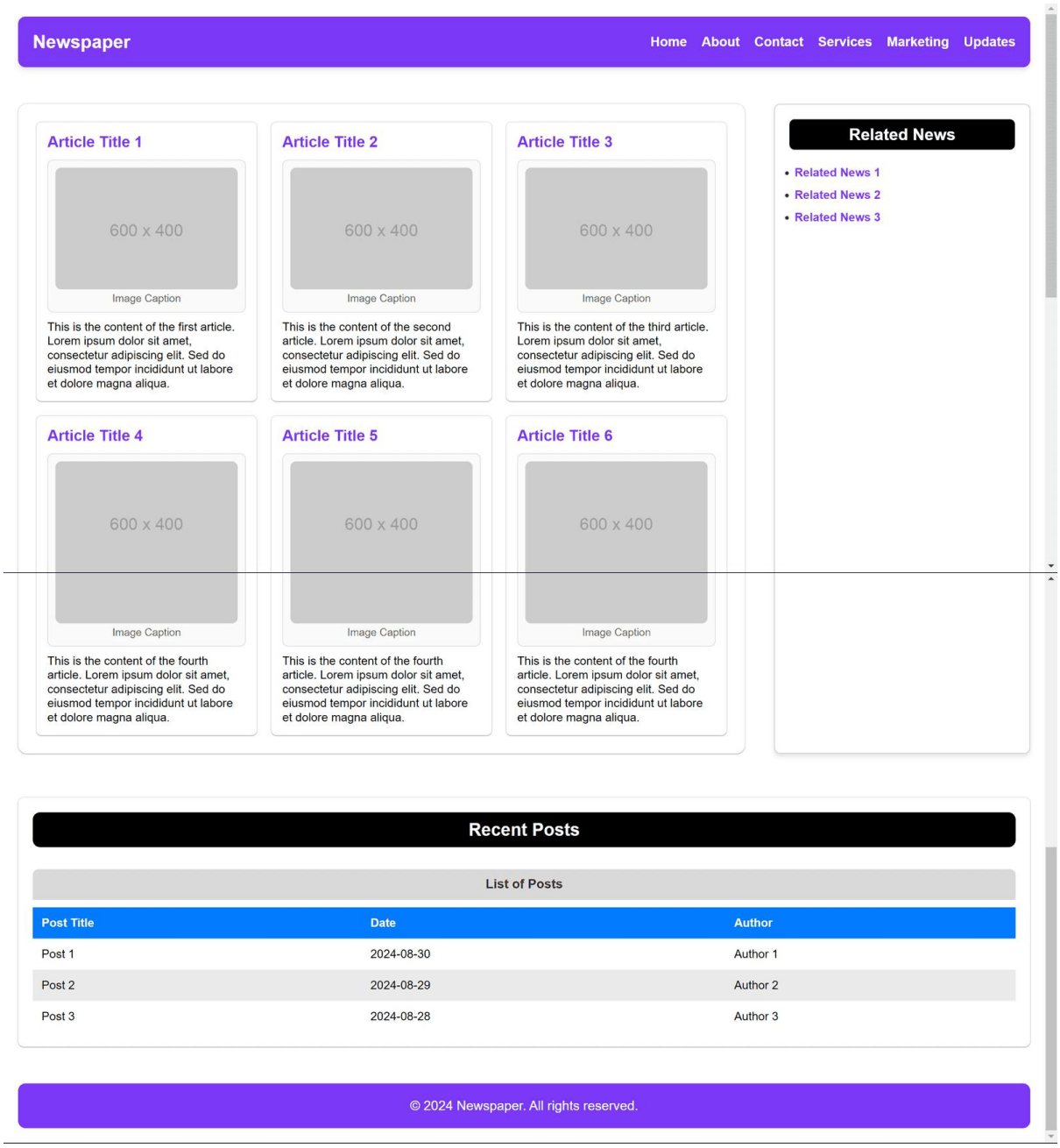
```
        }
        tbody tr:nth-child(even) {
            background-color: #f9f9f9;
        }
        tbody tr:hover {
            background-color: #eaeaea;
        }
        @media (max-width: 600px) {
            th,
            td {
                padding: 8px;
                font-size: 14px;
            }
        }
        caption {
            background-color: #d9d9d9;
            padding: 10px;
            font-weight: bold;
            border-bottom: 2px solid #ddd;
            border-radius: 8px 8px 0 0;
            font-size: 1.1em;
            color: #333;
        }
        section {
            margin-top: 40px;
            margin-bottom: 50px;
        }
        article h2 {
            color: #7b38f7;
            font-size: 1.3em;
            margin-bottom: 12px;
        }
        article p {
            text-align: left;
            line-height: 1.2;
            margin-top: 10px;
        }
        article:hover {
            background-color: #e7ddfb;
        }
        @media (max-width: 768px) {
            .content {
                flex-direction: column;
                padding: 10px;
            }
            aside {
                width: 100%;
                margin-top: 20px;
                position: static;
                margin-left: 0;
            }
            .main-content {
                grid-template-columns: 1fr;
            }
        }
    </style>
</head>
<body>
    <header>
        <a href="#">Newspaper</a>
        <nav>
            <a href="#">Home</a>
```

```
            <a href="#">About</a>
            <a href="#">Contact</a>
            <a href="#">Services</a>
            <a href="#">Marketing</a>
            <a href="#">Updates</a>
        </nav>
    </header>

    <div class="content">
        <main class="main-content">
            <article>
                <h2>Article Title 1</h2>
                <figure>
                    <img src="https://via.placeholder.com/600x400" alt="Placeholder Image">
                    <figcaption>Image Caption</figcaption>
                </figure>
                <p>This is the content of the first article. Lorem ipsum dolor sit amet, consectetur adipiscing elit.
                    Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.</p>
            </article>
            <article>
                <h2>Article Title 2</h2>
                <figure>
                    <img src="https://via.placeholder.com/600x400" alt="Placeholder Image">
                    <figcaption>Image Caption</figcaption>
                </figure>
                <p>This is the content of the second article. Lorem ipsum dolor sit amet, consectetur adipiscing elit.
                    Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.</p>
            </article>
            <article>
                <h2>Article Title 3</h2>
                <figure>
                    <img src="https://via.placeholder.com/600x400" alt="Placeholder Image">
                    <figcaption>Image Caption</figcaption>
                </figure>
                <p>This is the content of the third article. Lorem ipsum dolor sit amet, consectetur adipiscing elit.
                    Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.</p>
            </article>
            <article>
                <h2>Article Title 4</h2>
                <figure>
                    <img src="https://via.placeholder.com/600x400" alt="Placeholder Image">
                    <figcaption>Image Caption</figcaption>
                </figure>
                <p>This is the content of the fourth article. Lorem ipsum dolor sit amet, consectetur adipiscing elit.
                    Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.</p>
            </article>
            <article>
                <h2>Article Title 5</h2>
                <figure>
                    <img src="https://via.placeholder.com/600x400" alt="Placeholder Image">
                    <figcaption>Image Caption</figcaption>
                </figure>
                <p>This is the content of the fourth article. Lorem ipsum dolor sit amet, consectetur adipiscing elit.
                    Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.</p>
            </article>
            <article>
                <h2>Article Title 6</h2>
                <figure>
                    <img src="https://via.placeholder.com/600x400" alt="Placeholder Image">
                    <figcaption>Image Caption</figcaption>
                </figure>
                <p>This is the content of the fourth article. Lorem ipsum dolor sit amet, consectetur adipiscing elit.
```

```
                    Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.</p>
            </article>
        </main>
        <aside class="related-news">
            <h3>Related News</h3>
            <ul>
                <li><a href="#">Related News 1</a></li>
                <li><a href="#">Related News 2</a></li>
                <li><a href="#">Related News 3</a></li>
            </ul>
        </aside>
    </div>
    <section>
        <h2>Recent Posts</h2>
        <div>
            <table>
                <caption>List of Posts</caption>
                <thead>
                    <tr>
                        <th>Post Title</th>
                        <th>Date</th>
                        <th>Author</th>
                    </tr>
                </thead>
                <tbody>
                    <tr>
                        <td>Post 1</td>
                        <td>2024-08-30</td>
                        <td>Author 1</td>
                    </tr>
                    <tr>
                        <td>Post 2</td>
                        <td>2024-08-29</td>
                        <td>Author 2</td>
                    </tr>
                    <tr>
                        <td>Post 3</td>
                        <td>2024-08-28</td>
                        <td>Author 3</td>
                    </tr>
                </tbody>
            </table>
        </div>
    </section>
    <footer>
        <p>© 2024 Newspaper. All rights reserved.</p>
    </footer>
</body>
</html>
```

OUTPUT:


Newspaper
Home About Contact Services Marketing Updates
Article Title 1
600 x 400
Image Caption
This is the content of the first article. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.
Article Title 2
600 x 400
Image Caption
This is the content of the second article. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.
Article Title 3
600 x 400
Image Caption
This is the content of the third article. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.
Article Title 4
600 x 400
Image Caption
This is the content of the fourth article. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.
Article Title 5
600 x 400
Image Caption
This is the content of the fourth article. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.
Article Title 6
600 x 400
Image Caption
This is the content of the fourth article. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.
Related News
Related News 1
Related News 2
Related News 3
Recent Posts
List of Posts
Post Title
Date
Author
Post 1
2024-08-30
Author 1
Post 2
2024-08-29
Author 2
Post 3
2024-08-28
Author 3
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Program 6:

Apply HTML, CSS and JavaScript to design a simple calculator to perform the following operations: sum, product, difference, remainder, quotient, power, square-root and square.

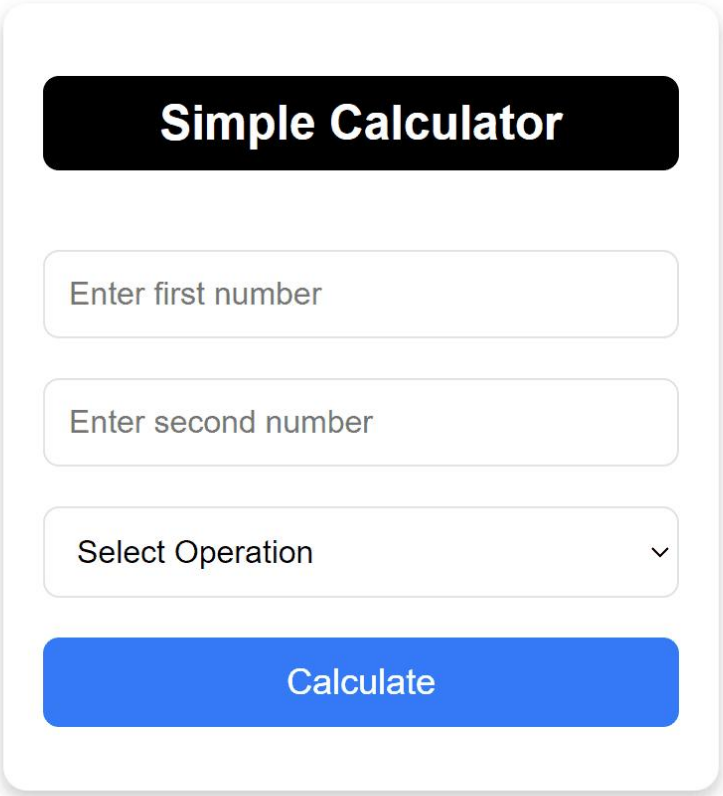
```
<!DOCTYPE html>
<head>
    <title>Simple Calculator | vtucode</title>
    <style>
        body {
            font-family: 'Arial', sans-serif;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
            margin: 0;
        }
        .calculator {
            background: #fff;
            padding: 20px;
            border-radius: 12px;
            box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
            width: 320px;
            text-align: center;
        }
        h1 {
            border-radius: 8px;
            background: #000;
            font-size: 24px;
            padding: 10px;
            color: #ffffff;
            margin-bottom: 30px;
        }
        input,
        select,
        button {
            width: 100%;
            margin: 10px 0;
            padding: 12px;
            border: 1px solid #0808081d;
            border-radius: 8px;
            font-size: 16px;
            box-sizing: border-box;
            transition: border-color 0.3s, box-shadow 0.3s;
        }
        #operation {
            cursor: pointer;
        }
        input:focus,
        select:focus,
        button:focus {
            outline: none;
            border-color: #007bff;
            box-shadow: 0 0 0 3px rgba(38, 143, 255, 0.25);
        }
        option {
            background-color: #fff;
            color: #000;
            padding: 10px;
            border: none;
        }
        option:hover {
```

```
            background-color: #f1f1f1;
        }
        button {
            background-color: #007bff;
            color: white;
            border: none;
            cursor: pointer;
            font-size: 18px;
            transition: box-shadow 0.3s, transform 0.3s;
        }
        button:hover {
            box-shadow: 0 0 0 2px #fff, 0 0 0 4px #007bff;
        }
        button:focus {
            box-shadow: 0 0 0 2px #fff, 0 0 0 4px #007bff;
        }
        #result.error {
            background: #ffdddd;
            border-color: #ff0000;
        }
        #result.success {
            font-size: 17px;
            font-weight: 500;
            color: #000;
            background: #6ef08d38;
            border-color: #47e56d;
        }
        #result {
            font-size: 18px;
            color: #000000;
            border-radius: 8px;
            background: #afffe2;
            border: 1px solid #ccc;
            box-shadow: 0 2px 4px rgba(0, 0, 0, 0.1);
            transition: opacity 0.5s, transform 0.5s;
            opacity: 0;
            transform: translateY(-20px);
        }
        #result.show {
            cursor: not-allowed;
            opacity: 1;
            margin-top: 20px;
            padding: 15px;
            transform: translateY(0);
        }
    </style>
</head>
<body>
    <div class="calculator">
        <h1>Simple Calculator</h1>
        <form id="calculator-form">
            <input type="number" id="num1" placeholder="Enter first number" required>
            <input type="number" id="num2" placeholder="Enter second number" required>
            <select id="operation" required>
                <option value="">Select Operation</option>
                <option value="sum">Sum</option>
                <option value="product">Product</option>
                <option value="difference">Difference</option>
                <option value="remainder">Remainder</option>
                <option value="quotient">Quotient</option>
                <option value="power">Power</option>
                <option value="sqrt">Square Root</option>
```

```
            <option value="square">Square</option>
        </select>
        <button type="button" onclick="calculate()">Calculate</button>
    </form>
    <div id="result"></div>
</div>
<script>
    function calculate() {
        const num1 = parseFloat(document.getElementById('num1').value);
        const num2 = parseFloat(document.getElementById('num2').value);
        const operation = document.getElementById('operation').value;
        let result = '';
        let resultClass = 'success';
        if (isNaN(num1) || isNaN(num2)) {
            result = 'Please enter valid numbers.';
            resultClass = 'error';
        } else {
            switch (operation) {
                case 'sum':
                    result = `Sum: ${num1 + num2}`;
                    break;
                case 'product':
                    result = `Product: ${num1 * num2}`;
                    break;
                case 'difference':
                    result = `Difference: ${num1 - num2}`;
                    break;
                case 'remainder':
                    result = `Remainder: ${num1 % num2}`;
                    break;
                case 'quotient':
                    if (num2 === 0) {
                        result = 'Cannot divide by zero';
                        resultClass = 'error';
                    } else {
                        result = `Quotient: ${num1 / num2}`;
                    }
                    break;
                case 'power':
                    result = `Power: ${Math.pow(num1, num2)}`;
                    break;
                case 'sqrt':
                    if (num1 < 0 || num2 < 0) {
                        result = 'Square root is not defined for negative numbers';
                        resultClass = 'error';
                    } else {
                        result = `Square Root of ${num1}: ${Math.sqrt(num1)} <br> Square Root of ${num2}:
${Math.sqrt(num2)}`;
                    }
                    break;
                case 'square':
                    result = `Square of ${num1}: ${Math.pow(num1, 2)} <br> Square of ${num2}: ${Math.pow(num2, 2)}`;
                    break;
                default:
                    result = 'Please select an operation.';
                    resultClass = 'error';
            }
        }
        const resultDiv = document.getElementById('result');
        resultDiv.innerHTML = result;
        resultDiv.classList.remove('show', 'error', 'success');
        resultDiv.classList.add(resultClass, 'show');
```

```
        }
    </script>
</body>
</html>
```

Output:



Program 7.

Develop JavaScript program (with HTML/CSS) for:
a) Converting JSON text to JavaScript Object
b) Convert JSON results into a date
c) Converting From JSON To CSV and CSV to JSON
d) Create hash from string using crypto.createHash() method

```
<!DOCTYPE html>
<head>
    <script src="https://cdnjs.cloudflare.com/ajax/libs/crypto-js/4.1.1/crypto-js.min.js"></script>
    <title>Simple Converter | vtucode</title>
    <style>
        * {
            padding: 0;
            margin: 0;
            box-sizing: border-box;
        }
        body {
            font-family: Arial, sans-serif;
            color: #000000;
        }
        .container {
            width: 60%;
            margin: 0 auto;
            padding: 20px;
        }
        .head-title h1 {
            font-size: 28px;
            padding: 10px;
            color: #fff;
            margin-bottom: 50px;
        }
        .head-title {
            width: 100%;
            background: #000;
            text-align: center;
            border-radius: 10px;
        }
        .section {
            margin-bottom: 40px;
            padding: 20px;
            border-radius: 8px;
            background: #fff;
            box-shadow: rgba(0, 0, 0, 0.1) 0px 1px 3px 0px, rgba(0, 0, 0, 0.06) 0px 1px 2px 0px;
            transition: all 0.3s;
            overflow: hidden;
        }
        .section h2 {
            color: #000000;
            font-size: 20px;
            margin-bottom: 15px;
        }
        textarea {
            font-size: 14px;
            width: 100%;
            height: 120px;
            margin-bottom: 15px;
            padding: 12px;
            border-radius: 8px;
```

```
        border: 1px solid #00000022;
        box-sizing: border-box;
        transition: border-color 0.3s ease, box-shadow 0.3s ease;
    }
    textarea:focus {
        background: transparent;
        border: 1px solid #00000022;
        border-color: #007BFF;
        box-shadow: 0 0 12px rgba(0, 123, 255, 0.5);
        outline: none;
    }
    input[type="text"] {
        width: calc(100% - 24px);
        padding: 12px;
        border-radius: 8px;
        border: 1px solid #ddd;
        box-sizing: border-box;
        transition: border-color 0.3s ease, box-shadow 0.3s ease;
        margin-bottom: 15px;
    }
    input[type="text"]:focus {
        border-color: #007BFF;
        box-shadow: 0 0 8px rgba(0, 123, 255, 0.5);
        outline: none;
    }
    button {
        display: inline-block;
        padding: 15px 15px;
        margin: 10px 0;
        font-weight: 600;
        border: none;
        border-radius: 7px;
        background-color: #007BFF;
        color: #fff;
        cursor: pointer;
        font-size: 16px;
        transition: box-shadow 0.3s ease, transform 0.3s ease;
    }
    button:hover {
        box-shadow: 0 0 0 2px #fff, 0 0 0 4px #007BFF;
    }
    button:focus {
        box-shadow: 0 0 0 2px #fff, 0 0 0 4px #007BFF;
    }
    pre {
        display: none;
        background: #f8f9fa;
        border: 1px solid #ddd;
        padding: 15px;
        border-radius: 8px;
        overflow: auto;
        transition: opacity 0.3s ease;
    }
    .error {
        margin-top: 10px;
        font-size: 14px;
        color: #000;
        background: #ffdddd;
        border-color: #ff0000;
        padding: 10px;
    }
    .success {
```

```
            margin-top: 10px;
            font-size: 14px;
            color: #000;
            background: #6ef08d38;
            border-color: #47e56d;
            padding: 10px;
        }
        .adjust-area {
            margin-top: 30px;
        }
    </style>
</head>
<body>
    <div class="container">
        <div class="head-title">
            <h1>Simple Converter</h1>
        </div>
        <div class="section">
            <h2>1. Convert JSON Text to JavaScript Object</h2>
            <textarea id="jsonInput" placeholder="Enter JSON here..."></textarea>
            <button onclick="convertJsonToObject()">Convert JSON</button>
            <pre id="jsonOutput" class="output"></pre>
        </div>
        <div class="section">
            <h2>2. Convert JSON Results into Date</h2>
            <textarea id="jsonDateInput" placeholder='Enter JSON with date in "yyyy-mm-dd" format'></textarea>
            <button onclick="convertJsonToDate()">Convert to Date</button>
            <pre id="jsonDateOutput" class="output"></pre>
        </div>
        <div class="section">
            <h2>3. Convert JSON to CSV and CSV to JSON</h2>
            <textarea id="jsonCsvInput" placeholder="Enter JSON for CSV conversion..."></textarea>
            <button onclick="convertJsonToCsv()">JSON to CSV</button>
            <pre id="csvOutput" class="output"></pre>
            <textarea id="csvInput" placeholder="Enter CSV here..." class="adjust-area"></textarea>
            <button onclick="convertCsvToJson()">CSV to JSON</button>
            <pre id="jsonCsvOutput" class="output"></pre>
        </div>
        <div class="section">
            <h2>4. Create Hash from String</h2>
            <input type="text" id="hashInput" placeholder="Enter string to hash">
            <button onclick="createHash()">Create Hash</button>
            <pre id="hashOutput" class="output"></pre>
        </div>
    </div>
    <script>
        function showResult(id, text, isSuccess) {
            const element = document.getElementById(id);
            element.textContent = text;
            element.className = isSuccess ? 'success' : 'error';
            element.style.display = 'block';
            element.style.opacity = '1';
        }
        function convertJsonToObject() {
            const jsonInput = document.getElementById('jsonInput').value;
            try {
                const jsonObject = JSON.parse(jsonInput);
                showResult('jsonOutput', JSON.stringify(jsonObject, null, 2), true);
            } catch (error) {
                showResult('jsonOutput', 'Invalid JSON', false);
            }
        }
```

```
function convertJsonToDate() {
    const jsonDateInput = document.getElementById('jsonDateInput').value;
    try {
        const data = JSON.parse(jsonDateInput);
        if (data.date && !isNaN(new Date(data.date).getTime())) {
            const date = new Date(data.date);
            showResult('jsonDateOutput', date.toString(), true);
        } else {
            showResult('jsonDateOutput', 'Invalid Date Format', false);
        }
    } catch (error) {
        showResult('jsonDateOutput', 'Invalid JSON', false);
    }
}
function convertJsonToCsv() {
    const jsonInput = document.getElementById('jsonCsvInput').value;
    try {
        const jsonArray = JSON.parse(jsonInput);
        if (Array.isArray(jsonArray) && jsonArray.length > 0) {
            const keys = Object.keys(jsonArray[0]);
            const csv = [
                keys.join(','),
                ...jsonArray.map(row => keys.map(key => JSON.stringify(row[key])).join(','))
            ].join('\n');
            showResult('csvOutput', csv, true);
        } else {
            showResult('csvOutput', 'Invalid JSON: Expected an array with objects.', false);
        }
    } catch (error) {
        showResult('csvOutput', 'Invalid JSON', false);
    }
}
function convertCsvToJson() {
    const csvInput = document.getElementById('csvInput').value;
    try {
        const lines = csvInput.trim().split('\n');
        if (lines.length > 1) {
            const keys = lines[0].split(',');
            if (keys.length > 0) {
                const jsonArray = lines.slice(1).map(line => {
                    const values = line.split(',');
                    return keys.reduce((obj, key, index) => {
                        obj[key] = values[index];
                        return obj;
                    }, {});
                });
                showResult('jsonCsvOutput', JSON.stringify(jsonArray, null, 2), true);
            } else {
                showResult('jsonCsvOutput', 'Invalid CSV: No columns found.', false);
            }
        } else {
            showResult('jsonCsvOutput', 'Invalid CSV: No data found.', false);
        }
    } catch (error) {
        showResult('jsonCsvOutput', 'Invalid CSV', false);
    }
}
function createHash() {
    const hashInput = document.getElementById('hashInput').value.trim();
    if (hashInput.length > 0) {
        const hash = CryptoJS.SHA256(hashInput).toString();
        showResult('hashOutput', hash, true);
```

```
            } else {
                showResult('hashOutput', 'Input cannot be empty', false);
            }
        }
    </script>
</body>
</html>
```

Output:

Convert JSON Text to JavaScript Object**********

```
{
   "name": "Alice",
   "age": 30,
   "city": "New York"
}
```

Convert JSON Results into Date**********

```
{
   "date": "2024-09-01"
}
```

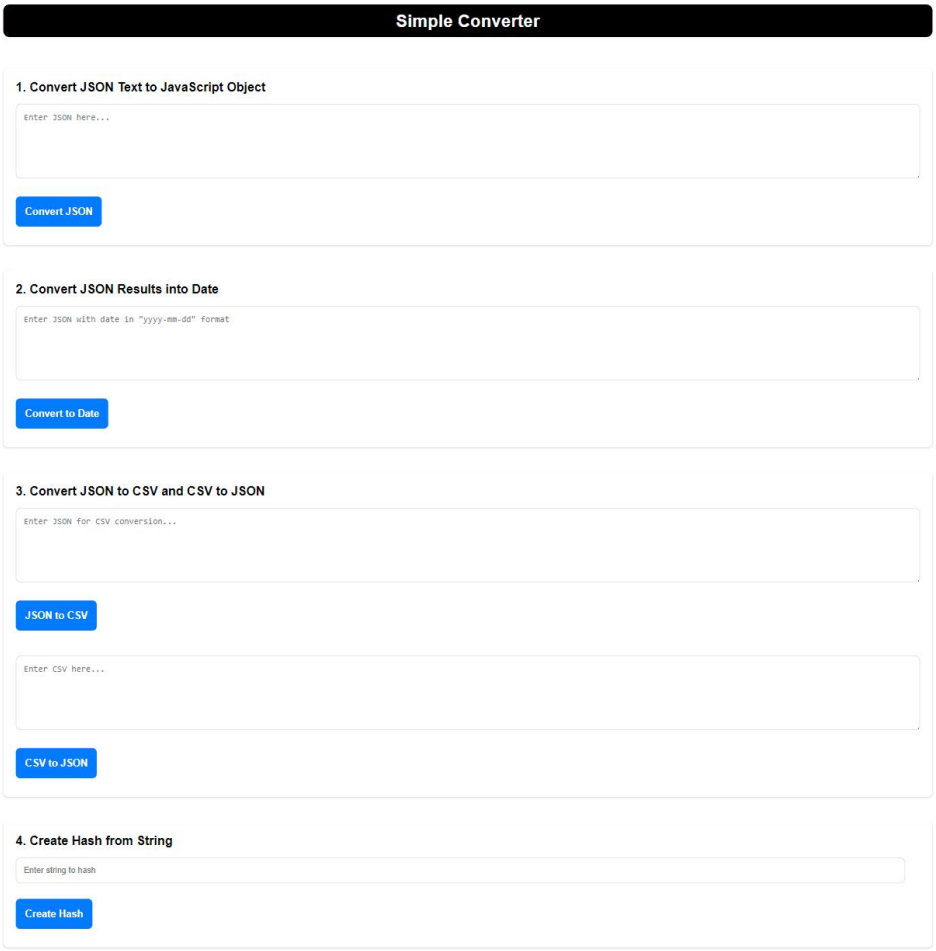
Convert JSON to CSV**********

```
[
   {"name": "Alice", "age": 30, "city": "New York"},
   {"name": "Bob", "age": 25, "city": "San Francisco"},
   {"name": "Charlie", "age": 35, "city": "Chicago"}
]
```

Convert CSV to JSON**********

```
name,age,city
Alice,30,New York
Bob,25,San Francisco
Charlie,35,Chicago
```

Create Hash from String**********

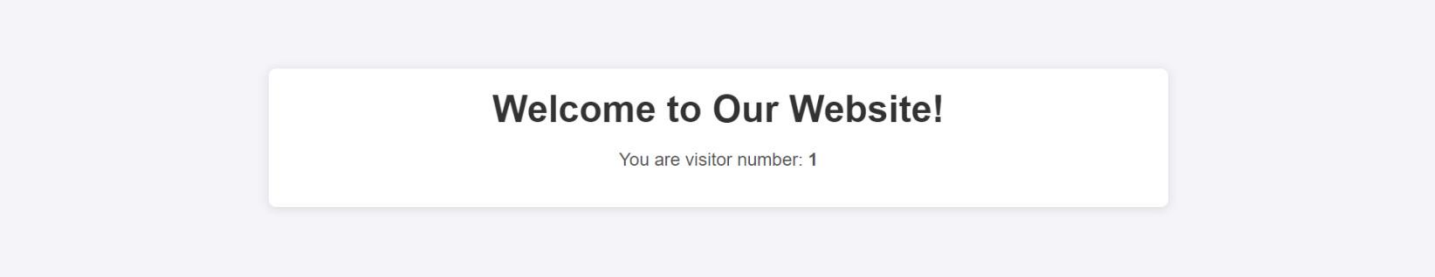

Simple Converter
1. Convert JSON Text to JavaScript Object
Enter JSON here...
Convert JSON
2. Convert JSON Results into Date
Enter JSON with date in "yyyy-mm-dd" format
Convert to Date
3. Convert JSON to CSV and CSV to JSON
Enter JSON for CSV conversion...
JSON to CSV
Enter CSV here...
CSV to JSON
4. Create Hash from String
Enter string to hash
Create Hash

Program 8:

a.Develop a PHP program (with HTML/CSS) to keep track of the number of visitors visiting the web page and to display this count of visitors, with relevant headings.

```php
<?php
$counterFile = "counter.txt";
if (!file_exists($counterFile)) {
    file_put_contents($counterFile, "0");
}
$currentCount = file_get_contents($counterFile);
$newCount = $currentCount + 1;
file_put_contents($counterFile, $newCount);
?>
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Visitor Counter | vtucode</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            text-align: center;
            margin: 0;
            padding: 0;
            display: flex;
            flex-direction: column;
            justify-content: center;
            height: 100vh;
            background-color: #f4f4f9;
            color: #333;
        }
        .container {
            background: #fff;
            padding: 20px;
            box-shadow: 0 2px 10px rgba(0, 0, 0, 0.1);
            border-radius: 8px;
            margin: 0 auto;
            width: 60%;
        }
        h1 {
            font-size: 2.5em;
            margin: 0;
        }
        p {
            font-size: 1.2em;
            color: #555;
        }
    </style>
</head>
<body>
    <div class="container">
        <h1>Welcome to Our Website!</h1>
        <p>You are visitor number: <strong><?php echo $newCount; ?></strong></p>
    </div>
</body>
</html>
```

Output:



b. Develop a PHP program (with HTML/CSS) to sort the student records which are stored in the database using selection sort.

```php
<?php
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "students";
$conn = new mysqli($servername, $username, $password, $dbname);
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
$sql = "SELECT * FROM students";
$result = $conn->query($sql);
$students = [];
if ($result->num_rows > 0) {
    while ($row = $result->fetch_assoc()) {
        $students[] = $row;
    }
}
function selectionSort(&$arr, $key)
{
    $n = count($arr);
    for ($i = 0; $i < $n - 1; $i++) {
        $minIndex = $i;
        for ($j = $i + 1; $j < $n; $j++) {
            if ($arr[$j][$key] < $arr[$minIndex][$key]) {
                $minIndex = $j;
            }
        }
        $temp = $arr[$i];
        $arr[$i] = $arr[$minIndex];
        $arr[$minIndex] = $temp;
    }
}
selectionSort($students, 'name');
?>
<!DOCTYPE html>
<head>
    <title>Sorted Student Records | vtucode</title>
    <style>
        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            background-color: #f0f2f5;
            color: #333;
            margin: 0;
            padding: 20px;
```

```
        }
        h2 {
            text-align: center;
            color: #4A90E2;
            margin-bottom: 20px;
        }
        table {
            width: 100%;
            border-collapse: collapse;
            background-color: #fff;
            border-radius: 10px;
            overflow: hidden;
            box-shadow: 0 2px 8px rgba(0, 0, 0, 0.1);
            margin: 0 auto;
        }
        th,
        td {
            padding: 12px 15px;
            text-align: left;
            border-bottom: 1px solid #ddd;
        }
        th {
            background-color: #4A90E2;
            color: white;
            text-transform: uppercase;
            letter-spacing: 0.03em;
        }
        tr {
            transition: background-color 0.3s ease;
        }
        tr:hover {
            background-color: #f1f1f1;
        }
        td {
            font-size: 0.9em;
            color: #555;
        }
        @media (max-width: 768px) {
            table,
            th,
            td {
                display: block;
                width: 100%;
            }
            th,
            td {
                box-sizing: border-box;
            }
            tr {
                margin-bottom: 15px;
                display: block;
                box-shadow: 0 2px 5px rgba(0, 0, 0, 0.1);
            }
            th {
                position: absolute;
                top: -9999px;
                left: -9999px;
            }
            td {
                border: none;
                position: relative;
                padding-left: 50%;
```

```
                text-align: right;
            }
            td:before {
                content: attr(data-label);
                position: absolute;
                left: 0;
                width: 50%;
                padding-left: 15px;
                font-weight: bold;
                text-align: left;
                text-transform: uppercase;
                color: #4A90E2;
            }
        }
    </style>
</head>
<body>
    <h2>Sorted Student Records by Name</h2>
    <table>
        <thead>
            <tr>
                <th>ID</th>
                <th>Name</th>
                <th>USN</th>
                <th>Branch</th>
                <th>Email</th>
                <th>Address</th>
            </tr>
        </thead>
        <tbody>
            <?php foreach ($students as $student): ?>
                <tr>
                    <td data-label="ID"><?php echo htmlspecialchars($student['id']); ?></td>
                    <td data-label="Name"><?php echo htmlspecialchars($student['name']); ?></td>
                    <td data-label="USN"><?php echo htmlspecialchars($student['usn']); ?></td>
                    <td data-label="Branch"><?php echo htmlspecialchars($student['branch']); ?></td>
                    <td data-label="Email"><?php echo htmlspecialchars($student['email']); ?></td>
                    <td data-label="Address"><?php echo htmlspecialchars($student['address']); ?></td>
                </tr>
            <?php endforeach; ?>
        </tbody>
    </table>
</body>
</html>
```

OUTPUT:

Sorted Student Records by Name

ID	NAME	USN	BRANCH	EMAIL	ADDRESS
2	Aman Kumar	1ME21CS002	CSE	aman@gmail.com	Chennai
8	Arun Kumar	1ME21CS008	CSE	arun@gmail.com	Bhopal
5	Bikash Kumar Singh	1ME21CS005	CSE	bikash@gmail.com	Mumbai, India
1	Braham Kumar	1ME21CS001	DS	braham@gmail.com	Nepal
9	Dipesh Kumar Mandal	1ME21CS009	ISE	dipesh@gmail.com	Indore
6	Shaoib Akhtar	1ME21CS006	AI&ML	shoaib@gmail.com	Patna, India
7	Shiv Kumar Yadav	1ME21CS007	ECE	shiv@gmail.com	Lucknow
4	Shubham Kumar	1ME21CS004	CSE	shubham@gmail.com	Ghaziabad
10	Shyam Kumar Singh	1ME21CS010	ME	shyam@gmail.com	Pune
3	Sunil Kumar	1ME21CS003	CIVIL	sunil@gmail.com	Delhi, India

Program 9:

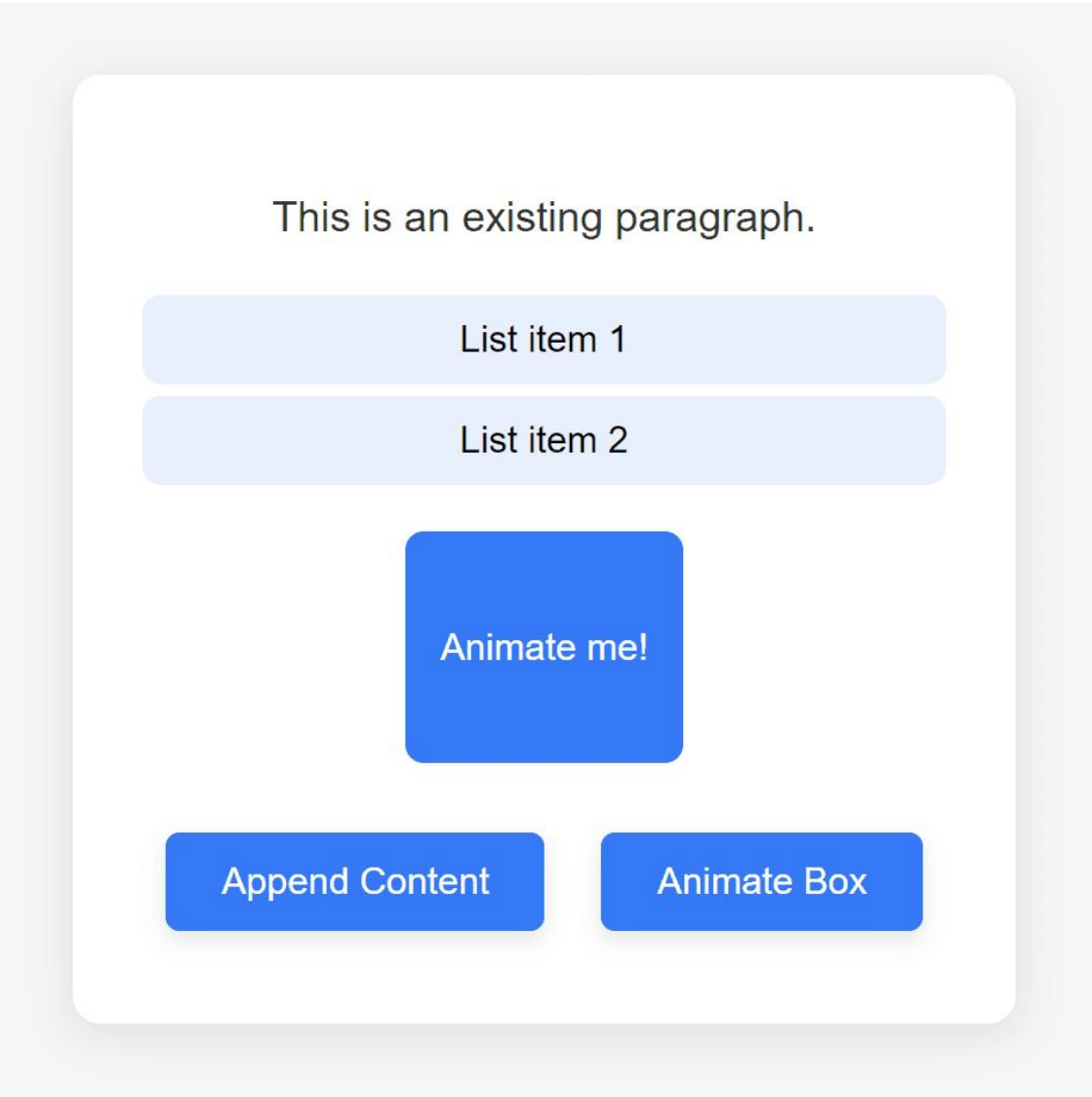
Develop jQuery script (with HTML/CSS) for:
a) Appends the content at the end of the existing paragraph and list.
b) Change the state of the element with CSS style using animate() method.
c) Change the color of any div that is animated.

```
<!DOCTYPE html>
<head>
    <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
    <title>jQuery Example | vtucode</title>
    <style>
        body {
            font-family: 'Roboto', sans-serif;
            background-color: #f4f7f6;
            margin: 0;
            padding: 0;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }
        .container {
            text-align: center;
            background: #fff;
            padding: 30px;
            border-radius: 12px;
            box-shadow: 0 4px 20px rgba(0, 0, 0, 0.1);
            transition: transform 0.3s ease-in-out;
        }
        .container:hover {
            transform: scale(1.02);
        }
        #paragraph {
            margin-bottom: 20px;
            color: #333;
            font-size: 18px;
            line-height: 1.5;
        }
        #list {
            margin-bottom: 20px;
            list-style: none;
            padding: 0;
        }
        #list li {
            background: #e8f0fe;
            margin: 5px 0;
            padding: 10px;
            border-radius: 8px;
            transition: background 0.3s;
        }
        #list li:hover {
            background: #d0e2fe;
        }
        .box {
            padding: 0 10px;
            width: 100px;
            height: 100px;
            background-color: #007bff;
            margin: 20px auto;
            line-height: 100px;
```

```
            color: white;
            text-align: center;
            border-radius: 8px;
            transition: all 0.3s ease;
        }
        button {
            padding: 12px 24px;
            margin: 10px;
            cursor: pointer;
            border: none;
            border-radius: 6px;
            font-size: 16px;
            background: #007bff;
            color: white;
            transition: box-shadow 0.3s, transform 0.2s;
            box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);
        }
        button:hover {
            box-shadow: 0 0 0 2px #fff, 0 0 0 4px #007bff;
        }
        button:focus {
            box-shadow: 0 0 0 2px #fff, 0 0 0 4px #007bff;
        }
        button:active {
            background: #004494;
            transform: translateY(0);
        }
    </style>
</head>
<body>
    <div class="container">
        <p id="paragraph">This is an existing paragraph.</p>
        <ul id="list">
            <li>List item 1</li>
            <li>List item 2</li>
        </ul>
        <div class="box" id="box">Animate me!</div>
        <button id="appendButton">Append Content</button>
        <button id="animateButton">Animate Box</button>
    </div>
    <script>
        $(document).ready(function () {
            $("#appendButton").click(function () {
                $("#paragraph").append(" Appended text.");
                $("#list").append("<li>New appended list item</li>");
            });
            $("#animateButton").click(function () {
                $("#box").stop(true, true).css({
                    width: "100px",
                    height: "100px",
                    opacity: 1,
                    backgroundColor: "blue"
                }).animate({
                    width: "200px",
                    height: "200px",
                    opacity: 0.5
                }, 1000, function () {
                    $(this).css("background-color", "green");
                });
            });
        });
    </script>
```

```
</body>
</html>
```

Output:



Program 10:

Develop a program that uses class Student which prompts the user to enter marks in three subjects and calculates total marks, percentage and displays the score card details. [Hint: Use list to store the marks in three subjects and total marks. Use __init__() method to initialize name, USN and the lists to store marks and total, Use getMarks() method to read marks into the list, and display() method to display the score card details.]

Textfile.txt

hi this is example text…

Data.json

{"name":"JohnDoe","age":30,"city":"NewYork","skills":["JavaScript","React","Node.js"],"address":{"street":"123ElmStreet","zipcode":"10001"},"projects":[{"name":"WebsiteRedesign","year":2023,"technologies":["HTML","CSS","JavaScript"]},{"name":"Mobile App","year":2024,"technologies":["React Native","Expo"]}]}

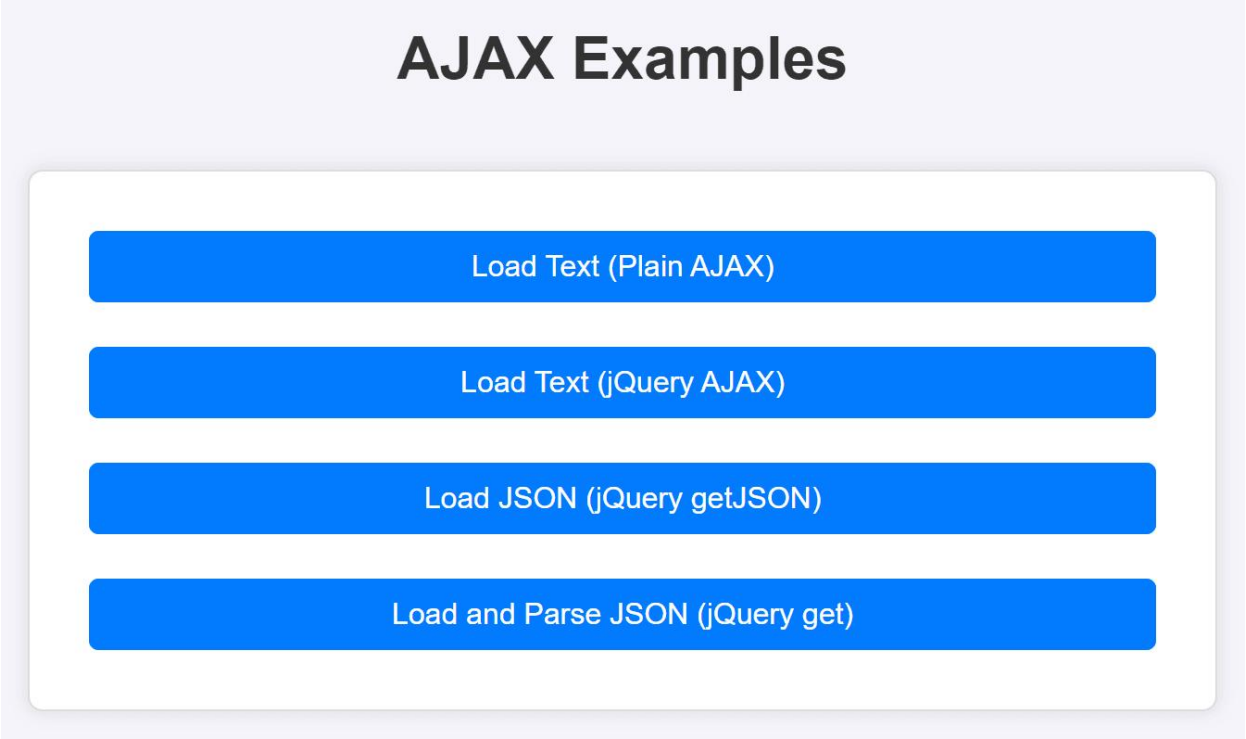
```
<!DOCTYPE html>
<head>
    <title>AJAX Examples | vtucode</title>
    <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.6.0/jquery.min.js"></script>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 0;
            padding: 0;
            background-color: #f4f4f9;
        }
        h1 {
            text-align: center;
            color: #333;
            padding: 20px 0;
        }
        #content {
            flex-direction: column;
            display: flex;
            max-width: 600px;
            margin: 20px auto;
            padding: 20px;
            border: 1px solid #ddd;
            border-radius: 8px;
            background-color: #fff;
            box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
        }
        button {
            display: inline-block;
            padding: 10px 15px;
            margin: 12px;
            border: none;
            border-radius: 5px;
            background-color: #007bff;
            color: #fff;
            font-size: 16px;
            cursor: pointer;
            transition: box-shadow 0.3s;
        }
        button:hover {
            box-shadow: 0 0 0 2px #fff, 0 0 0 4px #007bff;
        }
        button:focus {
```

```
            box-shadow: 0 0 0 2px #fff, 0 0 0 4px #007bff;
        }
        #output {
            display: none;
            margin-top: 20px;
            padding: 10px;
            border-radius: 5px;
            white-space: pre-wrap;
            max-height: 300px;
            overflow-y: auto;
        }
        #output.plain-ajax {
            background-color: #f0f8ff;
            border: 1px solid #b0c4de;
        }
        #output.jquery-ajax {
            background-color: #f5fffa;
            border: 1px solid #98fb98;
        }
        #output.jquery-json {
            background-color: #fffaf0;
            border: 1px solid #ffd700;
        }
        #output.parse-json {
            background-color: #fff0f5;
            border: 1px solid #ff69b4;
        }
    </style>
</head>
<body>
    <h1>AJAX Examples</h1>
    <div id="content">
        <button id="plain-ajax-btn">Load Text (Plain AJAX)</button>
        <button id="jquery-ajax-btn">Load Text (jQuery AJAX)</button>
        <button id="jquery-json-btn">Load JSON (jQuery getJSON)</button>
        <button id="parse-json-btn">Load and Parse JSON (jQuery get)</button>
        <div id="output"></div>
    </div>
    <script>
        function showOutput(className) {
            const output = document.getElementById('output');
            output.className = className;
            output.style.display = 'block';
        }
        document.getElementById('plain-ajax-btn').addEventListener('click', function () {
            var xhr = new XMLHttpRequest();
            xhr.open('GET', 'textfile.txt', true);
            xhr.onload = function () {
                if (xhr.status === 200) {
                    document.getElementById('output').innerText = xhr.responseText;
                } else {
                    document.getElementById('output').innerText = 'Error loading file.';
                }
                showOutput('plain-ajax');
            };
            xhr.send();
        });
        $('#jquery-ajax-btn').on('click', function () {
            $.ajax({
                url: 'textfile.txt',
                method: 'GET',
                success: function (data) {
```

```
                $('#output').text(data);
            },
            error: function () {
                $('#output').text('Error loading file.');
            }
        }).always(function () {
            showOutput('jquery-ajax');
        });
    });
    $('#jquery-json-btn').on('click', function () {
        $.getJSON('data.json')
            .done(function (data) {
                $('#output').text(JSON.stringify(data, null, 2));
            })
            .fail(function () {
                $('#output').text('Error loading JSON file.');
            })
            .always(function () {
                showOutput('jquery-json');
            });
    });
    $('#parse-json-btn').on('click', function () {
        $.get('data.json')
            .done(function (data) {
                try {
                    let jsonData;

                    if (typeof data === 'string') {
                        jsonData = JSON.parse(data);
                    } else {
                        jsonData = data;
                    }
                    $('#output').text(JSON.stringify(jsonData, null, 2));
                } catch (e) {
                    $('#output').text('Error parsing JSON: ' + e.message);
                }
            })
            .fail(function () {
                $('#output').text('Error loading JSON file.');
            })
            .always(function () {
                showOutput('parse-json');
            });
    });
  </script>
</body>
</html>
```

Output:


AJAX Examples
Load Text (Plain AJAX)
Load Text (jQuery AJAX)
Load JSON (jQuery getJSON)
Load and Parse JSON (jQuery get)